

SafePay MENA

Real-Time AI Telecom Fraud Shield for Instant Payments

Hackathon: GSMA MENA Ignite Hackathon 2026	Assigned Mentor: Eng. Abdullah A. Alkaoud (stc)
Theme: Theme 4 (Secure FinTech, Payments & Anti-Fraud)	Platforms: GSMA Open Gateway • Nokia NaC • Gemini 2.0 Flash
Developer: Karim Mohamed Abdelnabi (Solo Engineer)	Prototypes: Live REST/WebSocket Simulator (<10ms SLA)

The Instant Payment Boom Has a 3-Second Blind Spot

Settlement is instantaneous and irreversible. Recovery drops to near zero once funds move.

1.5 Billion Instant Transactions in Egypt (InstaPay 2024): EGP 2.9 Trillion volume.
Saudi Arabia Sarie & UAE Aani: Tens of thousands of instant push transfers daily.
The Dilemma: Payments settle in under 3 seconds with zero post-clearing recall window.

AED 4.99 Multiplier LexisNexis 2024	85% Scam Losses Vishing & Coercion	March 31, 2026 CBUAE Notice 2025/3057
UAE financial institutions lose AED 4.99 in total operational, legal, and reimbursement costs for every single dirham stolen by fraudsters.	The vast majority of instant fraud is Authorized Push Payment (APP) scams where victims are coerced on active phone calls.	Central Bank mandate strictly banning SMS OTPs for high-value payments and shifting 100% liability to banks.

The Real MENA Fraud Landscape: Two Distinct Vectors

Understanding how instant push rails differ from online e-commerce card checkout.

Attack Vector	Payment Channel	Vulnerability & Threat	SafePay Telecom Solution
Vector 1: Spam Call Vishing & Coercion 85% of regional losses	Instant Push Rails (InstaPay, Sarie, Aani)	InstaPay has NO SMS OTP for transfers; users enter IPN PIN. Scammers impersonate government/banks on active 20-min calls and coerce transfers.	CAMARA Scam Signal API: Detects active voice calls in real time. Prompts biometric Face ID challenge with anti-coercion modal.
Vector 2: Stolen Card CNP Fraud E-Commerce & 3DS	Online Checkouts & Wallet Cash-In	Compromised card credentials (PAN/CVV) drained via intercepted SMS OTPs or rogue devices. Banned under CBUAE Notice 2025/3057.	CAMARA Number Verification: Silent 3-legged cellular session match in 300ms. Eliminates SMS OTP code from the screen entirely.

The Regulatory Tsunami: 100% Liability Shift

Central banks are banning SMS OTPs and holding institutions financially responsible.

CBUAE Notice 2025/3057 (UAE)	SAMA Cybersecurity Framework (KSA)
• Complete phaseout of SMS & email OTPs by March 31, 2026. • 100% Liability Shift: Banks must fully reimburse customers if fraud occurs via SMS OTP. • Mandates carrier-grade silent possession authentication (CAMARA Number Verification).	• Strict Sarie 20,000 SAR instant limit with RTGS holding thresholds. • Mandated biometric SIM ownership caps (CST: 10 citizens, 2 expats). • Strict auditability rules requiring explainable AI for all automated transaction blocks.

The Institutional Reality: Banks can no longer blame customers for falling for phone scams or SMS phishing. Without real-time telecom network intelligence, banks bear 100% of the financial burden.

SafePay MENA: The AI Telecom-Banking Middleware

Bridging the intelligence gap between mobile networks and instant payment rails.

1. Silent Carrier Auth	2. Scam Signal Interruption	3. SIM Swap Freeze
Number Verification API: Silently confirms that the active cellular data session owns the SIM in 300ms. <i>Zero OTP on handset screen.</i>	Scam Signal API: Checks if phone is engaged in an active voice call. Prompts on-device Face ID with anti-coercion modal to break scammer hold.	SIM Swap Check API: Queries carrier core for SIM replacement in past 24-240 hours. Instantly freezes account takeover attempts.

Integration Position: SafePay acts as a synchronous pre-authorization hook in bank gateways. If risk score is <25, transaction clears in <200ms with zero friction. If 25-69, prompt Face ID biometric. If ≥70, block instantly.

Dual-Engine Architecture: Sub-10ms Fast Path + AI Audit

Decoupling instant mathematical authorization from asynchronous LLM compliance traces.

Component	1. Deterministic Risk Engine	2. Gemini 2.0 Flash Agent
Role & Execution	Synchronous inline authorization decision (APPROVE / STEP_UP / BLOCK)	Asynchronous natural language compliance & audit logger
Latency SLA	Sub-10ms SLA (Benchmark: 0.0024ms / eval)	Sub-500ms asynchronous background generation
Throughput	> 400,000 evaluations per second	Event-driven streaming via WebSocket / Supabase
Regulatory Value	Enforces SAMA 20k SAR limit & CBE 70k EGP cap instantly	Generates plain-language legal trace citing SAMA & CBUAE controls

Standardized GSMA CAMARA APIs & Zero PII Architecture

Leveraging Nokia Network as Code to query mobile cores across stc, e&, and Vodafone.

API Name	Type	Carrier Query	SafePay Fraud Role
Number Verification	3-Legged	Cellular Bearer Match	Replaces SMS OTP with silent carrier data-session auth in 300ms.
SIM Swap Check	2-Legged	HLR / HSS Timestamp	Checks if SIM was swapped in last 24-240h. Account takeover shield.
Scam Signal	2-Legged	Voice Call State	Detects active unverified voice calls. Intercepts vishing coercion.
Device Status / Roaming	2-Legged	VLR Roaming Country	Identifies unexpected foreign roaming and unreachable handsets.

Zero PII Principle: SafePay never passes bank account numbers, payee identities, or balances to carriers. Only E.164 MSISDN identifiers and boolean verification tokens are exchanged.

Live Demonstration: 3 Real-World Scenarios

Split-screen interactive prototype tested and verified with 100% test pass rate.

Scenario 1: Clean Transfer Routine Payment (InstaPay)	Scenario 2: Vishing Scam Active Call Coercion (Sarie)	Scenario 3: SIM Swap Takeover Account Hijack (Aani)
• Transfer: 200 EGP to Mother • Number Verified: PASS • SIM Swapped: NO • Active Call: NO • Risk Score: 8 / 100 • Outcome: APPROVE in 200ms	• Transfer: 15,000 SAR to Unknown • Scam Signal: ACTIVE CALL • High-Value Unserved Payee • Risk Score: 52 / 100 • Outcome: STEP-UP Face ID + Anti-Coercion Warning Banner	• Transfer: 35,000 SAR to Mule • SIM Swapped: YES (2.1h ago) • Device Match: FAIL • Exceeds SAMA 20k SAR Limit • Risk Score: 94 / 100 • Outcome: HARD BLOCK + SAMA Compliance Trace

Explainable AI Audit Trace: Gemini 2.0 Flash

Transforming microsecond signals into court-admissible regulatory protection.

Scenario 3 Compliance Trace Output (Citing SAMA & CBUAE):		
Regulatory Rule Citation: SAMA-AML-2023-SEC4.2 & CBUAE-NOTICE-2025-3057		
Primary Threat Vector: SIM_SWAP_ACCOUNT_TAKEOVER (Critical Tier 3)		
Mathematical Risk Score: 94 / 100 (Automated Block Enforced)		
Executive Rationale: <i>'Transaction of 35,000 SAR blocked. Amount exceeds SAMA Sarie 20,000 SAR instant threshold. Carrier telemetry from stc confirms subscriber SIM was reissued 2.1 hours ago. Device hardware IMEI mismatch detected. Automated block completely eliminates institutional liability under CBUAE Notice 2025/3057.'</i>		
Immediate Proof Gives fraud ops plain-language evidence to resolve customer complaints.	Zero Hallucinations Deterministic fallback ensures 100% uptime even if LLM rate-limited.	Immutable Storage Persisted in Supabase PostgreSQL with tamper-proof timestamps.

Business Model & Unit Economics: 693% Bank ROI

High-margin telecom API monetization aligned with massive bank fraud savings.

Financial Metric	Legacy Status Quo	With SafePay MENA
Cost per Stolen Currency Unit	AED 4.99 Lost per AED 1 Stolen (LexisNexis)	AED 0.00 (Fraud intercepted before settlement)
Authentication Cost	\$0.03 - \$0.05 per SMS OTP code	\$0.20 per evaluated transaction (replaces SMS)
Wholesale Telco API Cost	\$0.00 (Zero network monetization)	\$0.07 metered query paid to stc / e& / Vodafone
SafePay Gross Margin	N/A	65% Gross Software Margin (\$0.13 net per eval)
Bank ROI (Tier-1 Bank)	Negative (Escalating fraud losses)	693% First-Year Net ROI (\$8.5M savings vs \$1.2M cost)

Competitive Advantage: Why SafePay Wins

Uniting fragmented carriers and payment rails into a single regional middleware.

Dimension	Legacy Fraud (FICO / SAS)	Individual Telcos (stc alone)	SafePay MENA
Network Visibility	Zero (Blind to SIM & calls)	Single carrier only (no roaming)	Multi-carrier MENA aggregator (+20, +966, +971)
Vishing Protection	None (PIN entered by victim)	Raw Scam Signal boolean	Scam Signal + Biometric Anti-Coercion modal
Latency Performance	Post-transaction / Batch	Raw network API latency	Sub-10ms deterministic matrix (0.0024ms actual)
Regulatory Shield	Generic global alerts	None (Data vendor only)	SAMA 2026 & CBUAE Notice 2025/3057 mapping

Validation: stc Mentorship & Empirical Evidence

Grounding architecture in carrier realities and 299 authoritative research sources.

1. stc Mentorship Alignment	2. 299-Source Research Synthesis	3. Functional Prototype
• Session with Eng. Abdullah Alkaoud (stc Open Gateway Lead). • Validated 250ms circuit breaker SLA and carrier billing rules. • Pivoted to prioritize active vishing call detection (85% scam volume).	• Deep research notebook (299 sources). • Validated UAE AED 4.99 fraud multiplier. • Analyzed CBUAE Notice 2025/3057 & SAMA Counter-Fraud framework. • Verified 44% scam reduction in CAMARA pilots.	• FastAPI backend running on port 8000. • Split-screen Next.js/Tailwind simulation. • Nokia NaC SDK integration + fallback. • 100% automated test pass rate. • 0.0024ms benchmark evaluation speed.

Regulatory Compliance & Data Sovereignty Roadmap

Built from day one to satisfy central bank data residency and cybersecurity frameworks.

SAMA Cybersecurity Framework	CST SIM Ownership Caps	CBUAE Liability Shift
- Aligns with SAMA 4 domains & 96 controls. - Enforces Sarie 20k SAR instant limits. - Immutable Supabase cryptographic logs.	- Enforces Saudi CST biometric SIM caps. - Cross-checks multi-SIM velocity. - Flags unregistered roaming SIMs.	- Eliminates SMS OTP reliance before March 2026. - Shifts liability away from banks. - Silent 3-legged possession proof.

Sovereign Cloud Deployment: SafePay is packaged as containerized microservices ready for sovereign cloud hosting within Saudi Arabia (stc Cloud / Oracle Cloud Riyadh) and Egypt, satisfying strict cross-border data protection laws.

Commercial Go-To-Market & Scaled Expansion

Phased rollout across digital challenger banks, national switches, and retail giants.

Phase 1: Carrier Pilot (M1-M4)	Phase 2: GCC Commercial (M5-M12)	Phase 3: Scale to Egypt (Y2+)
• Sandboxed pilot with stc Open Gateway & STC Bank. • Pre-auth check on transfers >10k SAR. • Refine active vishing detection triggers.	• Commercial launch on Sarie & Aani. • Onboard top 5 GCC retail banks. • Target e-commerce 3DS checkouts ahead of March 2026 CBUAE deadline.	• Integrate with Egypt EBC / InstaPay (1.5B annual transactions). • Expand to Jordan, Qatar & Oman. • Enterprise revenue share with telcos.

SafePay MENA: Protecting the Payment Revolution

The bridge between \$275 billion in instant payments and the telecom network that can protect them.

*'Telecom networks possess the real-time truth.
Instant payment rails possess the settlement urgency.
Central bank regulations have set the deadline: March 31, 2026.
SafePay MENA is the operational bridge that unites them.'*

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Hackathon: GSMA MENA Ignite (Theme 4)	Live Prototype: http://127.0.0.1:8000 (Tested & Operational)
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